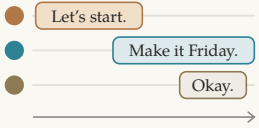
(a) Enrollment

recorded once by the user;
its first 5 s are used

in real use it may have:

- only 2–4 s recorded
- only 1 s of speech
- leading silence
- background noise
- reverberation
- another voice inside
- a very low level
- 4 kHz narrowband

(b) Several talkers



the user joins or leaves mid-clip;
1–4 others, turns and overlap

→ Make it Friday.

(c) Short segments

sounds like the user

VAD cuts of 1.2–6 s;
a similar voice, 0.5–1.6 s

→ *(empty)*

(d) No one speaking

song with vocals

noise, music, songs with vocals,
traffic, crowds, near silence

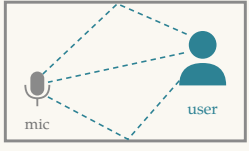
→ *(empty)*

(e) Noisy places

café, street, traffic, babble;
phone and studio recordings

→ I'm almost there.

(f) Far field, rooms



measured and simulated rooms;
enrollment near, mixture far

→ Close the window.

(g) Real recordings

meetings, phone calls and web
conversations, not mixed

→ *(empty)*

enrolled user · other people · similar voice · expected: only the user's words · expected: nothing